CHAOS 101

Activity Guide

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1. Introduction to Chaos

What is chaos?

Chaos bridges the gap between order and randomness. It is **deterministic yet unpredictable**: while it follows exact mathematical laws, its long-term behavior remains impossible to forecast.

It is not randomness — it is **extreme sensitivity to initial conditions**.

Applications and Importance of Chaos

Why study chaos?

- **Understanding Natural Systems:** From weather forecasting to stock market dynamics.
- **Cryptography and Security:** Utilizing extreme sensitivity to initial conditions to generate secure keys.
- **Biomedical Engineering:** Analyzing cardiac rhythms and brain activity patterns.

In our circuit, chaos enables **TRNG** by leveraging physical unpredictability over predictable algorithms. This ensures mathematically unpredictable sequences, greatly enhancing digital encryption and cybersecurity.

The Chua's circuit

This specific circuit was selected because it offers the simplest and most effective way to easily integrate and modulate complex chaotic behaviors in hardware..

Why Chua's circuit?

- Simplest system for real chaos
- Fully op-amp based (TL082)
- Iconic *double-scroll* attractor

2. The Device Interface

The Hardware

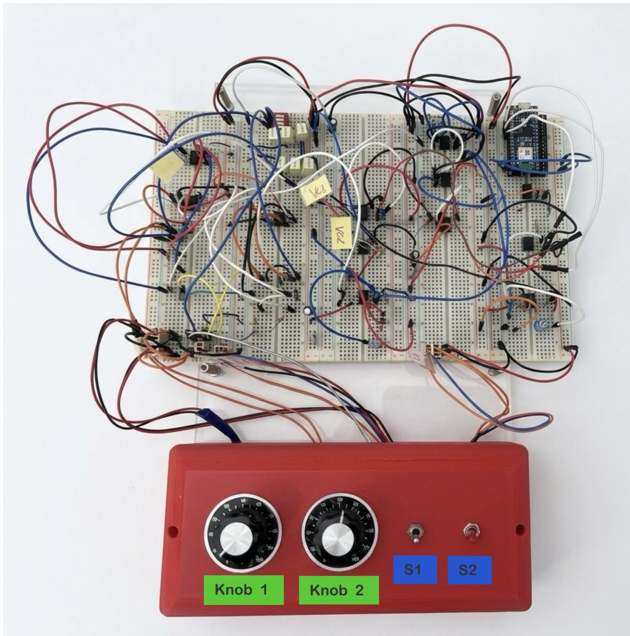


Figure 1: CHAOS 101 hardware station.

Control Interface

- **Knobs 1 & 2**: Potentiometers to vary resistance and modify the output.
- **Switch 1**: Master switch for the dual 9V battery supply.
- **Switch 2**: Activates the ESP32 for 10 kHz sampling.

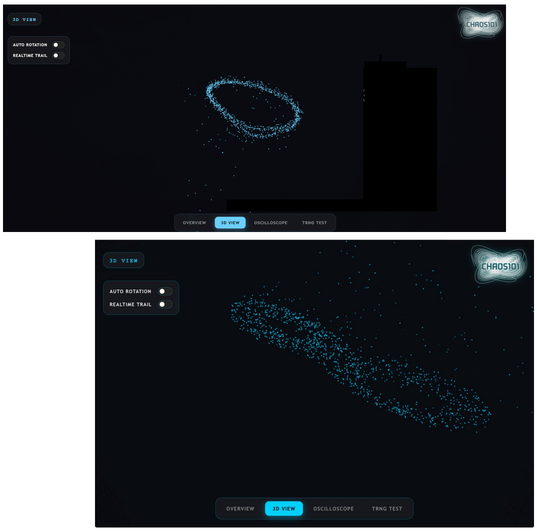
3. Activity: The First Flight

How to power the sistem?

1. Connect the hardware via the **USB-C** input.
2. Toggle **Switch 1** to power the circuit via batteries.
3. Toggle **Switch 2** to power the ESP32 and enable the 5V data link.

4. Web interface

Thanks to the onboard Wi-Fi powered by the Arduino-compatible ESP32, the CHAOS 101 interface is now live and accessible to everyone. Simply connect your own device to our web portal to explore the 'order within disorder' in real-time.

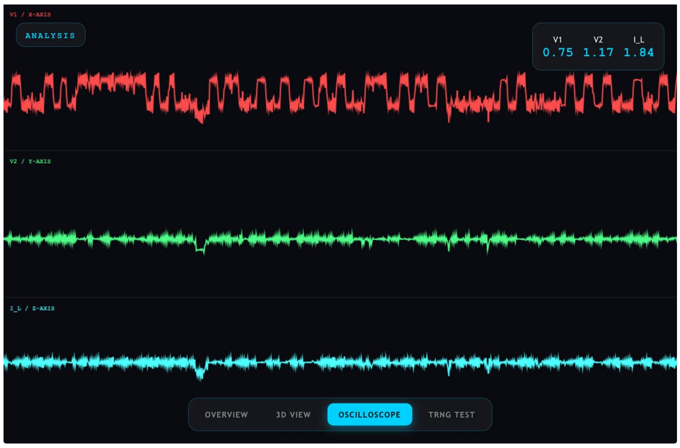


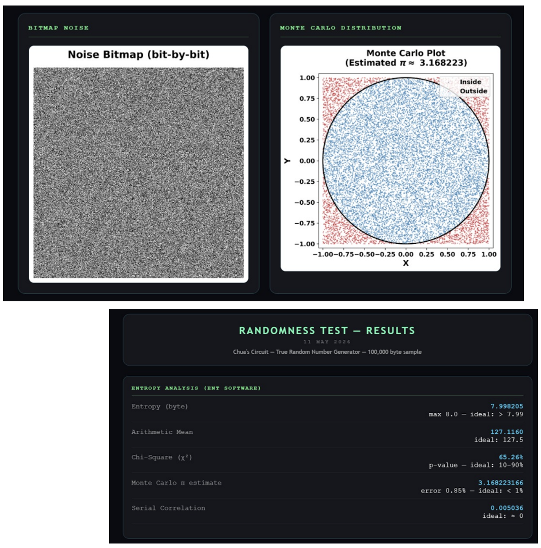
3D phase space

1. Turn **Knob 2** three full rotations to set the system's equilibrium point.
2. Observe the **periodic** regime, which appears on the display as a stable elliptical trajectory.
3. Slowly rotate **Knob 1** until the trajectories begin to intersect and diverge.
4. Continue adjusting until the **Double-Scroll** attractor forms—you have now generated real chaos.

Live oscilloscope

1. **Real-Time State Monitoring:** Monitor the circuit's core state variables (V_1, V_2, and I_L) directly in the time domain.
2. **Chaotic Fluctuations:** Witness unpredictable signals that never repeat the same pattern, illustrating the system's extreme sensitivity.





TRNG

1. **Analyze the TRNG Plots:** Observe and interpret the generated visual plots to understand the source behavior.
2. **Monte Carlo Plot:** Verify the geometric randomness via the π estimation (3.168223), checking that the error remains below 1%.
3. **Bitmap & Entropy:** Analyze the spatial independence on the Noise Bitmap and verify that the entropy is near the ideal maximum (7.998205 bits/byte).